

A Brief Introduction to Clustering

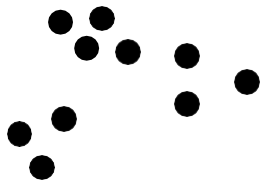
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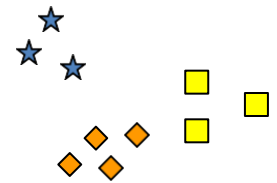
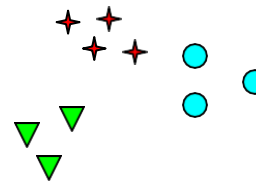
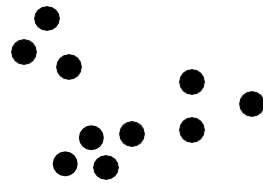
What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups.
 - Based on information found in the data that describes the objects and their relationships.
 - Also known as unsupervised classification.
- Many applications
 - Understanding: group related documents for browsing or to find genes and proteins that have similar functionality.
 - Summarization: Reduce the size of large data sets.

Notion of a Cluster is Ambiguous

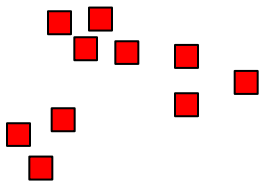


Initial points.

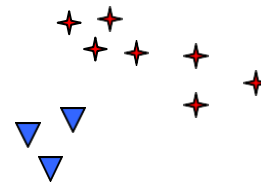
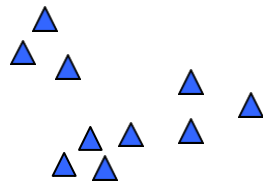


Six Clusters

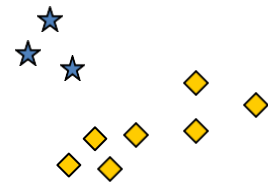
The notion of a cluster in cluster analysis is ambiguous due to several interrelated factors, which collectively mean there is no universal definition of what constitutes a "cluster."



Two Clusters



Four Clusters



Notion of a Cluster is Ambiguous

1) Algorithmic Diversity:

Different clustering algorithms impose distinct assumptions about cluster structure. For example:

1. *K-means* assumes spherical, equally sized clusters.
2. *DBSCAN* identifies dense regions separated by sparse areas.
3. *Hierarchical clustering* builds nested groupings based on proximity.

These varying assumptions lead to divergent results, as each algorithm "sees" clusters through its own lens.

2) Subjectivity in Criteria:

Clusters can be defined by density, connectivity, centroid proximity, or statistical distributions. Without a ground truth, the choice of criteria is often subjective.

Notion of a Cluster is Ambiguous

3) Data Characteristics:

- 1. Overlap/Ambiguity:** Data may lack clear separation, making boundaries between clusters fuzzy (e.g., overlapping Gaussian distributions).
- 2. High Dimensionality:** In high-dimensional spaces, distance metrics lose discriminative power (the "curse of dimensionality"), complicating cluster identification.
- 3. Noise/Outliers:** Some methods treat outliers as noise (e.g., DBSCAN), while others force them into clusters (e.g., k-means), altering interpretations.

4) Parameter Sensitivity:

Many algorithms require user-defined parameters (e.g., number of clusters k , density thresholds). Small changes in these parameters can yield vastly different results, and there is often no objective way to set them.

Notion of a Cluster is Ambiguous

3) Data Characteristics:

- 1. Overlap/Ambiguity:** Data may lack clear separation, making
bo Density thresholds are parameters used in **density-based**
dis **clustering algorithms** (e.g., DBSCAN) to define the minimum
conditions required for a region of data points to qualify as a
cluster. They typically involve two components:
 - **Distance threshold (ϵ):** The radius within which neighboring points are considered part of the same cluster.
- 2. High minimum density:**
 - **Minimum points (minPts):** The number of points required within the ϵ -radius to label a region as "dense."
- 3. Noisy data:**

4) Parameter Sensitivity:

Many algorithms require user-defined parameters (e.g., number of clusters k , density thresholds). Small changes in these parameters can yield vastly different results, and there is often no objective way to set them.

Notion of a Cluster is Ambiguous

5) Context-Dependent Goals:

The purpose of clustering (e.g., anomaly detection vs. customer segmentation) influences what qualifies as a meaningful cluster. For example, one application might prioritise small, dense clusters, while another seeks balanced groupings.

Example 1: Anomaly Detection

- **Goal:** Separate "normal" data (clusters) from rare outliers.
- **Cluster Criteria:** Focus on **large, dense regions** as normal. Treat small/sparse regions as anomalies.
- **Parameter Choice:** High *minPts* or large ϵ to ignore small groups.

Example 2: Customer Segmentation

- **Goal:** Identify distinct, interpretable groups for targeted marketing.
- **Cluster Criteria:** **Balanced cluster sizes** for actionable insights.
- **Parameter Choice:** Predefined k (number of segments) based on business needs.

Notion of a Cluster is Ambiguous

6) Validation Challenges:

Without labeled data, evaluating clusters relies on internal metrics (e.g., Silhouette Score, Davies-Bouldin index), which may conflict across algorithms. This lack of consensus on evaluation exacerbates ambiguity.

Types of Clusterings

- A *clustering* is a set of clusters.
- One crucial distinction is between hierarchical and partitional sets of clusters.

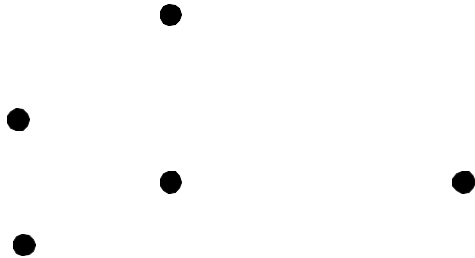
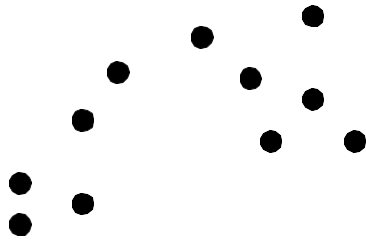
Partitional Clustering

- A division of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset.

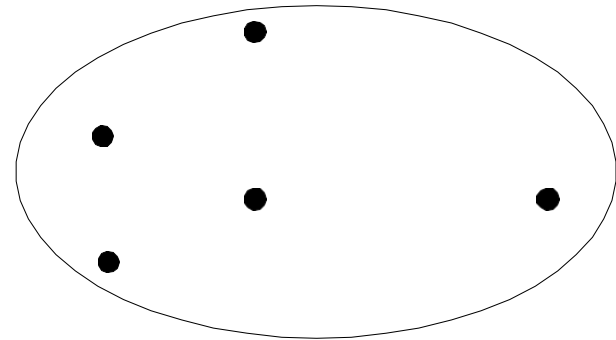
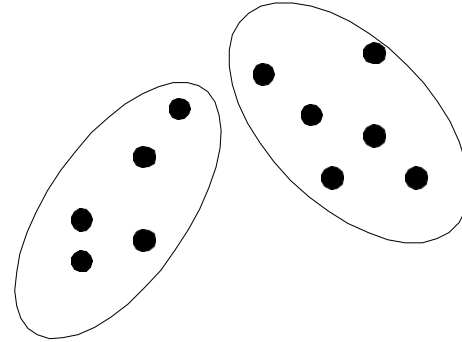
Hierarchical clustering

- A set of nested clusters organised as a hierarchical tree.

Partitional Clustering

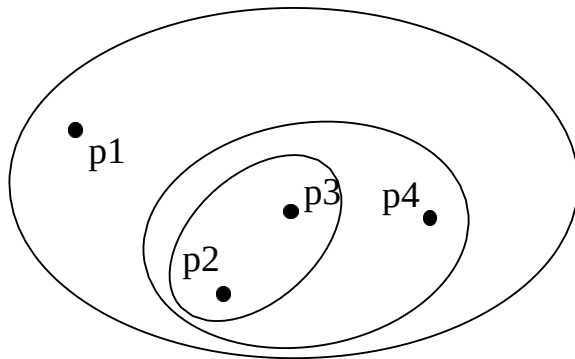


Original Points

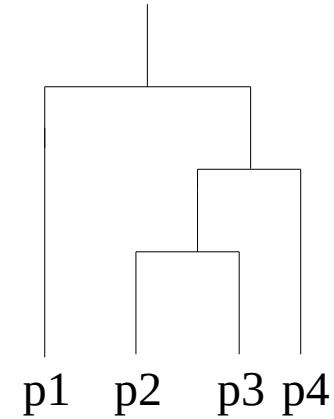


A Partitional Clustering

Hierarchical Clustering



Traditional Hierarchical Clustering



Traditional Dendrogram

Types of Clusterings

Other important distinctions are between:

- **Exclusive** versus **Overlapping** (non-exclusive) versus **Fuzzy** (probabilistic).
- **Complete** versus **Partial**.

- Exclusive: Each data point belongs to exactly one cluster. E.g., K-means, Hierarchical Clustering.
- Overlapping: Points can belong to multiple clusters (hard assignments). E.g., Overlapping community detection.
- Fuzzy (Probabilistic): Points have probabilistic memberships (e.g., 0.7 in Cluster A, 0.3 in Cluster B). E.g., Fuzzy C-means, Gaussian Mixture Models.
- Complete: All points are assigned to clusters (no unlabelled data). E.g., K-means, Agglomerative Clustering.
- Partial: Some points are excluded as noise/outliers. E.g., DBSCAN, HDBSCAN.

Types of Clusterings

Other important distinctions are between:

Why This Matters?

Application Alignment:

- Use *exclusive* clustering for customer segmentation (distinct groups).
- Use *fuzzy* clustering for text topics (documents can belong to multiple topics).
- Use *partial* clustering for anomaly detection (separate outliers).

Clustering is rarely one-size-fits-all—the choice depends on data structure, goals, and interpretability needs!

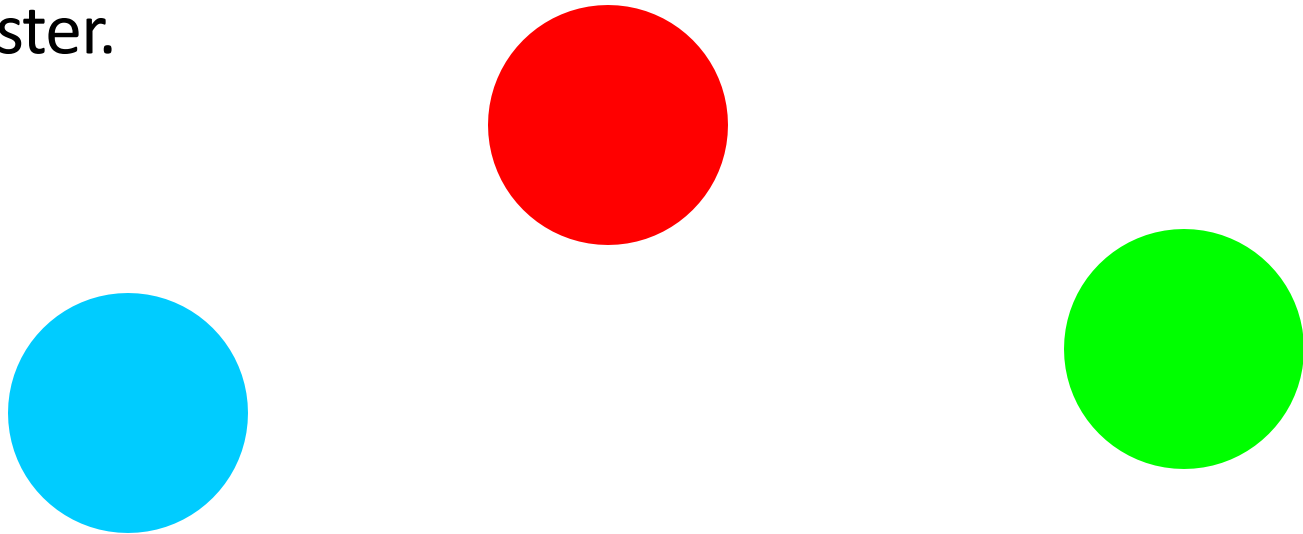
in Cluster B). E.g., Fuzzy C-means, Gaussian Mixture Models.

Complete: All points are assigned to clusters (no unlabelled data). E.g., K-means, Agglomerative Clustering.

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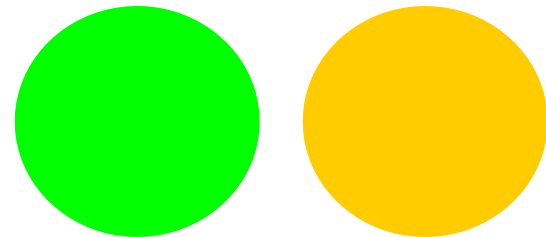
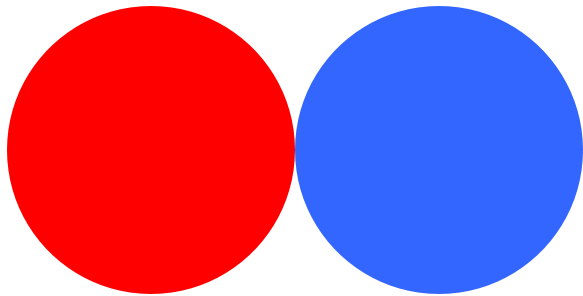
Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



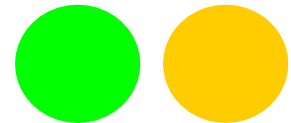
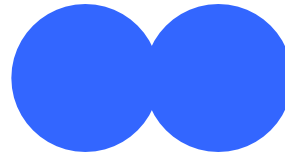
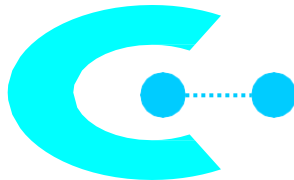
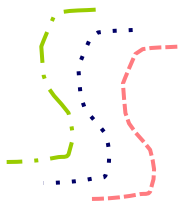
Types of Clusters: Center-Based

- Center-based
 - A cluster is a set of objects such that an object in a cluster is closer (more similar) to the “center” of a cluster, than to the center of any other cluster.
 - The center of a cluster is often a *centroid*, the average of all the points in the cluster, or a *medoid*, the most “representative” point of a cluster.



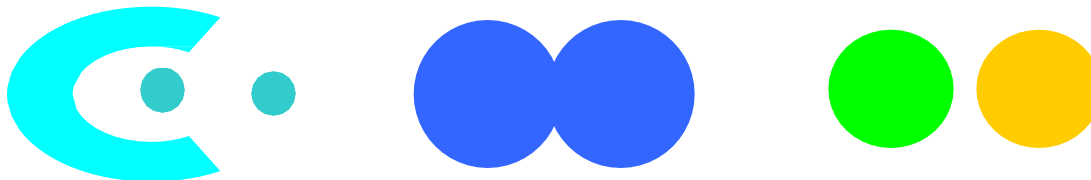
Types of Clusters: Contiguity-Based

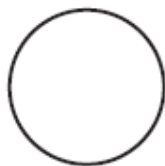
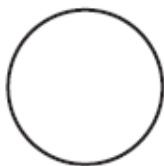
- Contiguous Cluster(Nearest neighbor or Transitive)
 - A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.



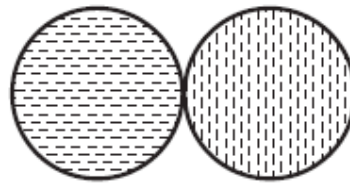
Types of Clusters: Density-Based

- Density-based
 - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
 - Used when the clusters are irregular or intertwined, and when noise and outliers are present.
 - The three curves don't form clusters since they fade into the noise, as does the bridge between the two small circular clusters.

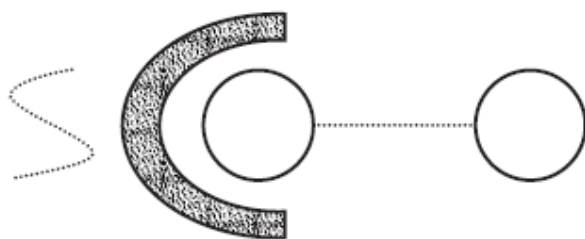




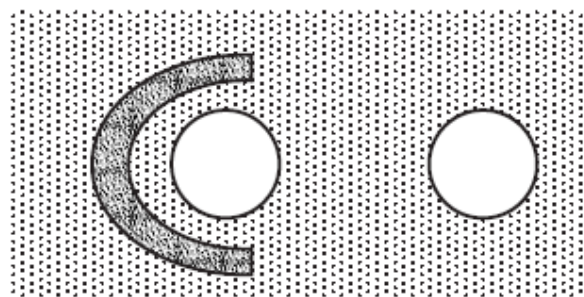
(a) Well-separated clusters. Each point is closer to all of the points in its cluster than to any point in another cluster.



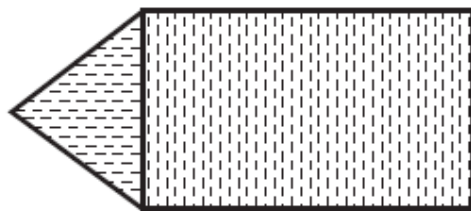
(b) Center-based clusters. Each point is closer to the center of its cluster than to the center of any other cluster.



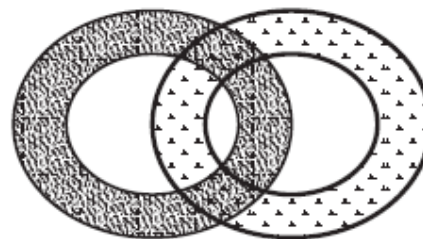
(c) Contiguity-based clusters. Each point is closer to at least one point in its cluster than to any point in another cluster.



(d) Density-based clusters. Clusters are regions of high density separated by regions of low density.



(e) Conceptual clusters. Points in a cluster share some general property that derives from the entire set of points. (Points in the intersection of the circles belong to both.)



Similarity and Dissimilarity

- Similarity
 - Numerical measure of how alike two data objects are.
 - Is higher when objects are more alike.
 - Often falls in the range $[0,1]$
- Dissimilarity
 - Numerical measure of how different two data objects are.
 - Is lower when objects are more alike.
 - Minimum dissimilarity is often 0.
 - Upper limit varies
- Proximity refers to a similarity or dissimilarity

Euclidean Distance

- Euclidean Distance

$$\mathbf{dist} = \sqrt{\sum_{k=1}^n (\mathbf{p}_k - \mathbf{q}_k)^2}$$

- where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .
- Standardization is necessary, if scales differ.

Common Properties of a Distance

- Distances, such as the Euclidean distance, have some well-known properties:
 1. $d(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ only if $p = q$.
(Positive definiteness)
 2. $d(p, q) = d(q, p)$ for all p and q . (Symmetry)
 3. $d(p, r) \leq d(p, q) + d(q, r)$ for all points p , q , and r .
(Triangle Inequality)
 - where $d(p, q)$ is the distance (dissimilarity) between points (data objects), p and q .
- A distance that satisfies these properties is a *metric*

Common Properties of a Similarity

- Similarities, also have some well-known properties:
 1. $s(p, q) = 1$ (or maximum similarity) only if $p = q$.
 2. $s(p, q) = s(q, p)$ for all p and q . (Symmetry)
 - where $s(p, q)$ is the similarity between points (data objects), p and q .

K-means Clustering

- Partitional clustering approach.
- Each cluster is associated with a *centroid* (center point).
- Each point is assigned to the cluster with the closest centroid.
- Number of clusters, K , must be specified.
- The basic algorithm is very simple.

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- ‘Closeness’ is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to ‘Until relatively few points change clusters’
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

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- K-means mentioned

- Most of t

- Often change

Suppose:

- $n = 10,000$ points,
- $K = 20$ clusters,
- $I = 15$ iterations,
- $d = 50$ attributes.

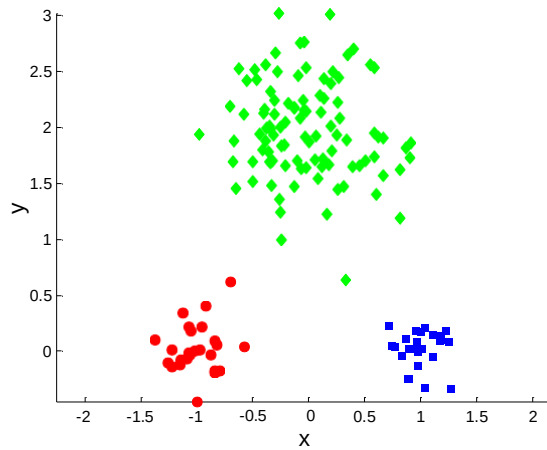
Total operations $\approx 10,000 \times 20 \times 15 \times 50 = 150,000,000$ operations.

- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

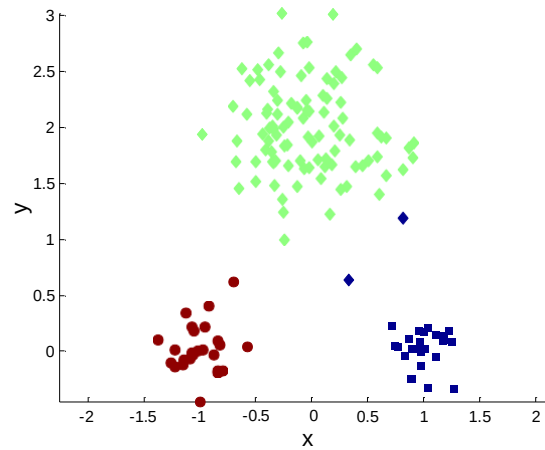
Evaluating K-means Clusters

- Most common measure is the *Sum of the Squared Error* (SSE)
 - For each point, the error is the distance to the nearest cluster.
 - To get SSE, we square these errors and sum them.
 - Given two clusters, we can choose the one with the smallest error.
 - One easy way to reduce SSE is to increase K, the number of clusters.
 - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K.

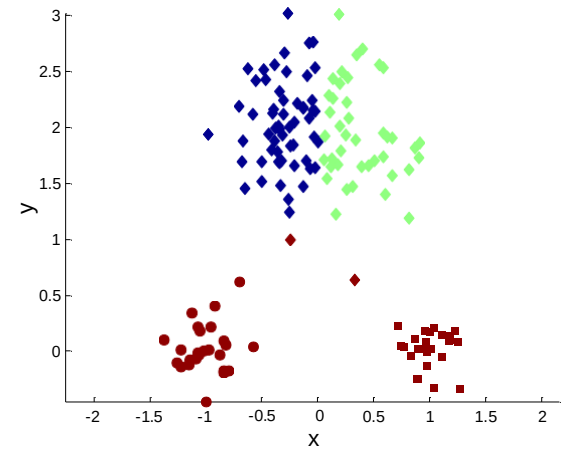
Two different K-means Clusterings



Original Points

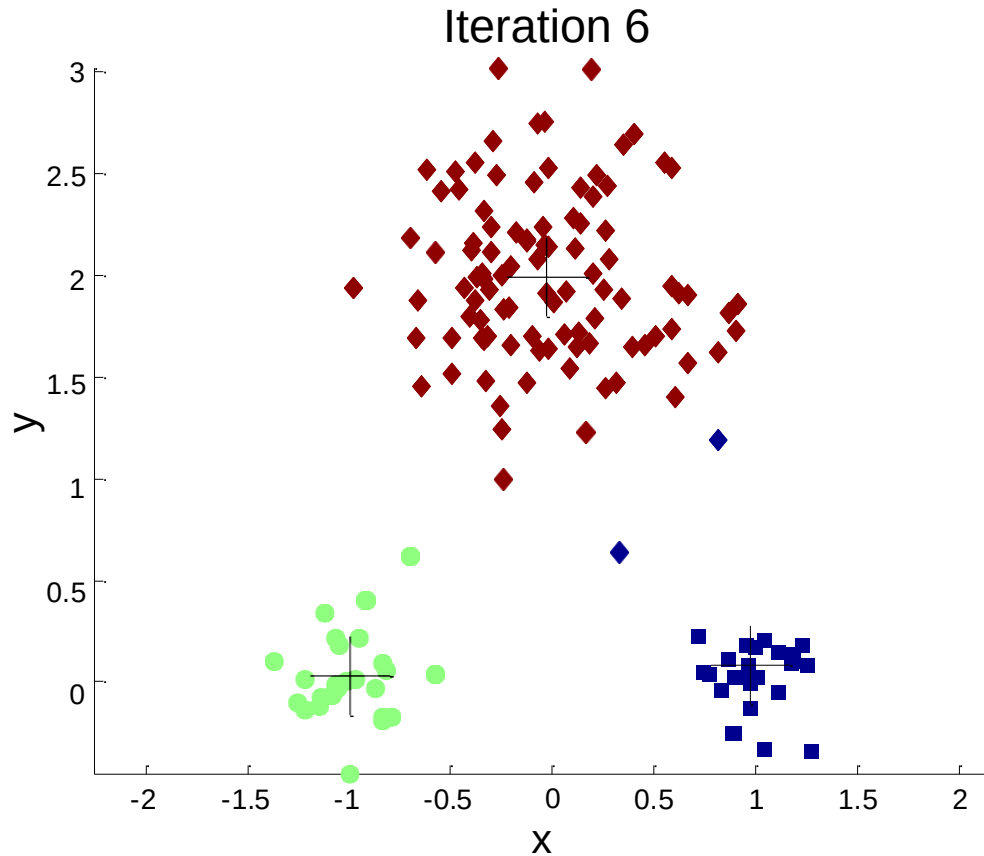


Optimal Clustering

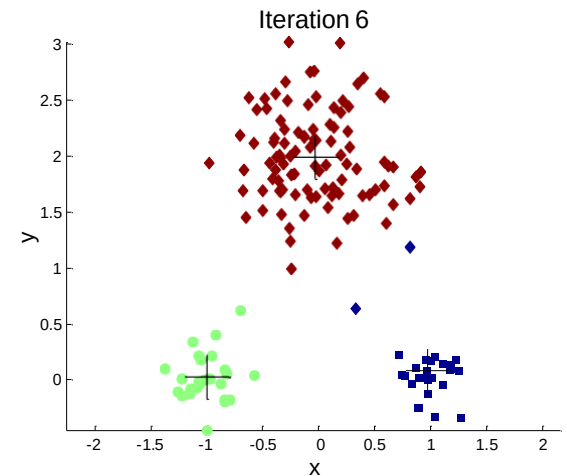
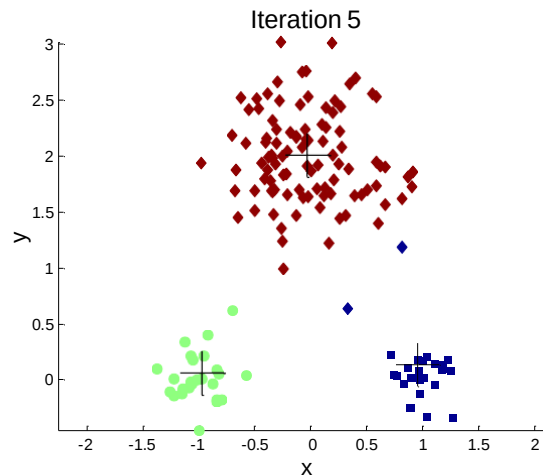
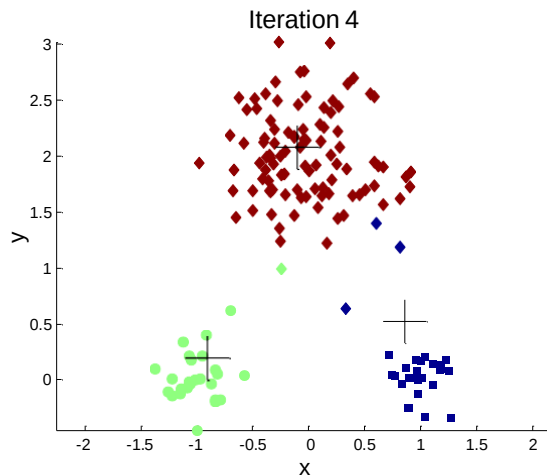
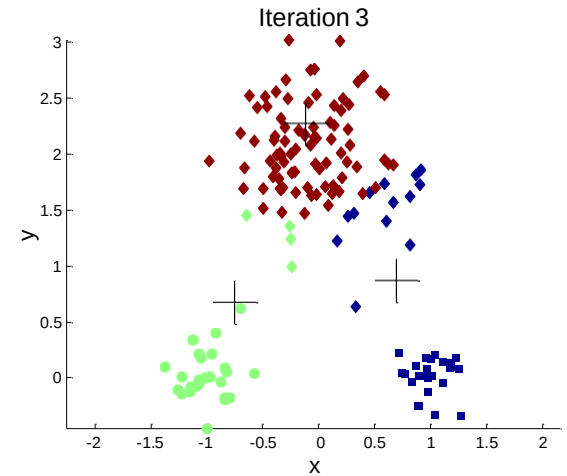
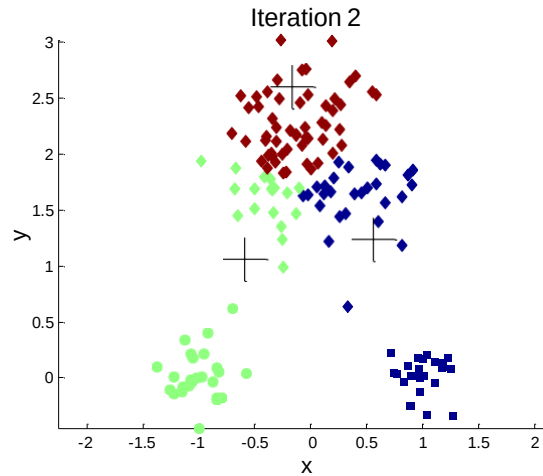
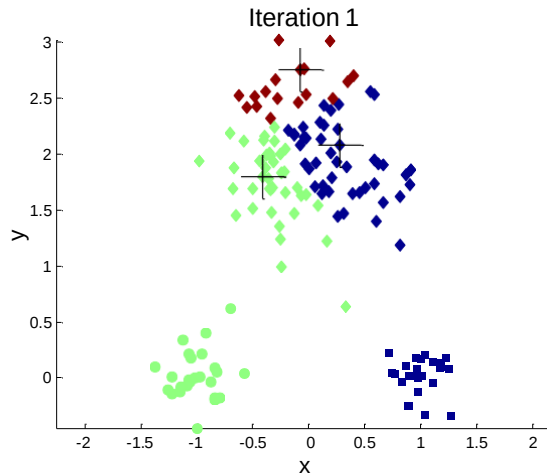


Sub-optimal Clustering

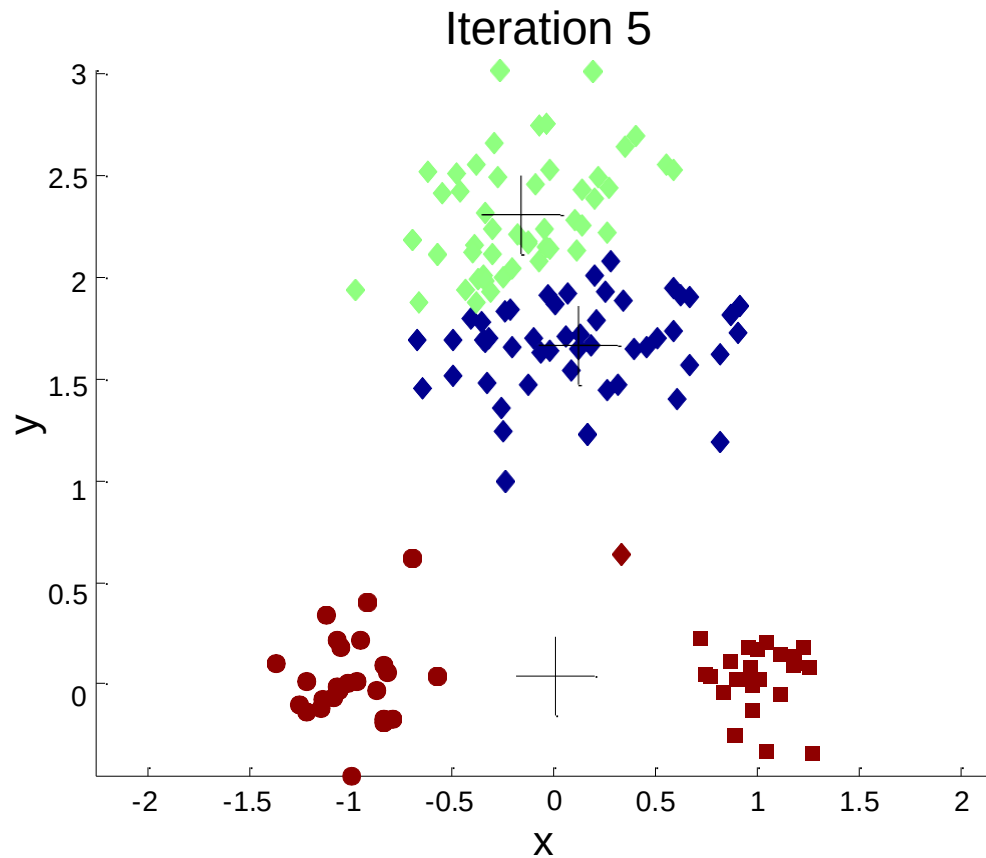
Importance of Choosing Initial Centroids



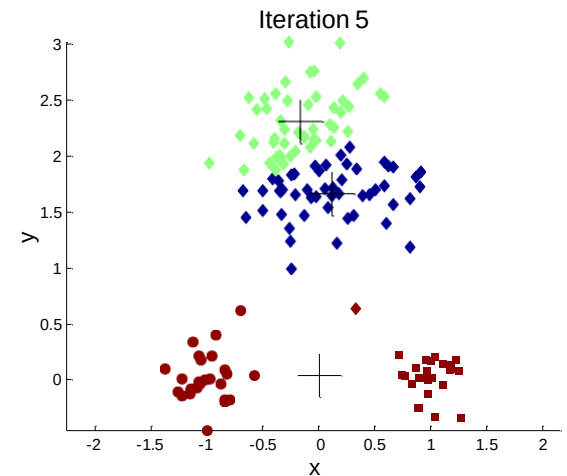
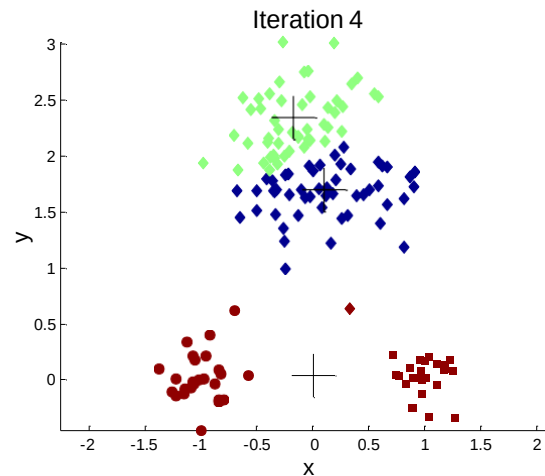
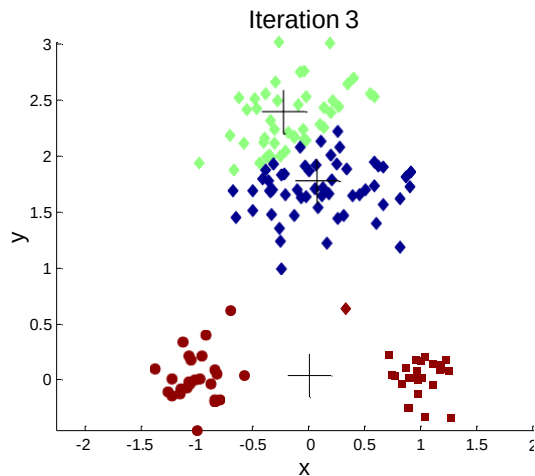
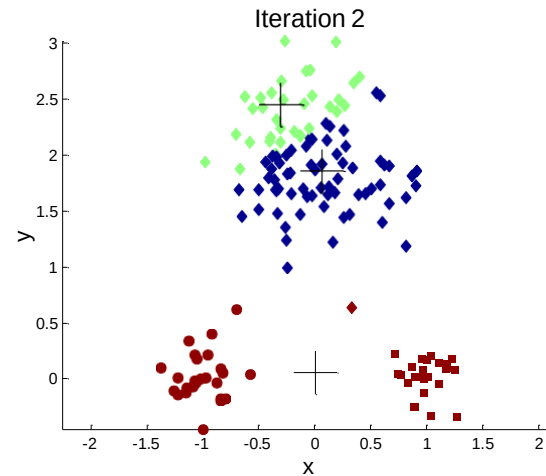
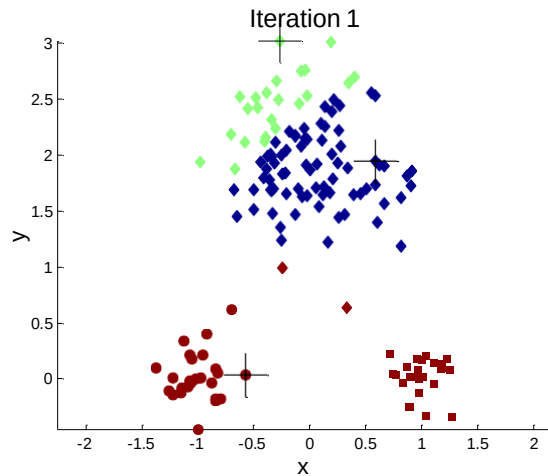
Importance of Choosing Initial Centroids



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Importance of Choosing Initial Centroids



Euclidean Distance

- Euclidean Distance

$$\mathbf{dist} = \sqrt{\sum_{k=1}^n (\mathbf{p}_k - \mathbf{q}_k)^2}$$

- where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .
- Standardization is necessary, if scales differ.

Minkowski Distance

- Minkowski Distance is a generalization of Euclidean Distance

$$\mathbf{dist} = \left(\sum_{k=1}^n | \mathbf{p}_k - \mathbf{q}_k |^r \right)^{\frac{1}{r}}$$

- where r is a parameter, n is the number of dimensions (attributes) and p_k and q_k are, respectively, the k^{th} attributes (components) or data objects p and q .

Minkowski Distance: Examples

- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.
 - A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors.
- $r = 2$. Euclidean distance.
- $r \rightarrow \infty$. “supremum” (L_{max} norm, L_∞ norm) distance.
 - This is the maximum difference between any component of the vectors.
- Do not confuse r with n , i.e., all these distances are defined for all numbers of dimensions.

Similarity Between Binary Vectors

- Common situation is that objects, p and q , have only binary attributes.
- Compute similarities using the following quantities
 - M_{01} = the number of attributes where p was 0 and q was 1
 - M_{10} = the number of attributes where p was 1 and q was 0
 - M_{00} = the number of attributes where p was 0 and q was 0
 - M_{11} = the number of attributes where p was 1 and q was 1
- Simple Matching and Jaccard Coefficients
- $SMC = \text{number of matches} / \text{number of attributes}$
$$= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$$
- $J = \text{number of } 11 \text{ matches} / \text{number of not-both-zero attributes values}$
$$= (M_{11}) / (M_{01} + M_{10} + M_{11})$$

SMC versus Jaccard: Example

$p = 1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$

$q = 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 1$

$M_{01} = 2$ (the number of attributes where p was 0 and q was 1)

$M_{10} = 1$ (the number of attributes where p was 1 and q was 0)

$M_{00} = 7$ (the number of attributes where p was 0 and q was 0)

$M_{11} = 0$ (the number of attributes where p was 1 and q was 1)

$$\begin{aligned}\text{SMC} &= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00}) \\ &= (0 + 7) / (2 + 1 + 0 + 7) = 0.7\end{aligned}$$

$$\begin{aligned}J &= (M_{11}) / (M_{01} + M_{10} + M_{11}) \\ &= 0 / (2 + 1 + 0) = 0\end{aligned}$$

Cosine Similarity

- If d_1 and d_2 are two document vectors,
then $\cos(d_1, d_2) = \frac{(d_1 \bullet d_2)}{||d_1|| ||d_2||}$,

where $\bullet$ indicates vector dot product and $||d||$ is the length of vector d .

- Example:

$$d_1 = \mathbf{3\ 2\ 0\ 5\ 0\ 0\ 0\ 2\ 0\ 0}$$

$$d_2 = \mathbf{1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 2}$$

$$d_1 \bullet d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$$

$$||d_1|| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} = 6.481$$

$$||d_2|| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2)^{0.5} = (6)^{0.5} = 2.245$$

$$\cos(d_1, d_2) = .3150$$

Correlation

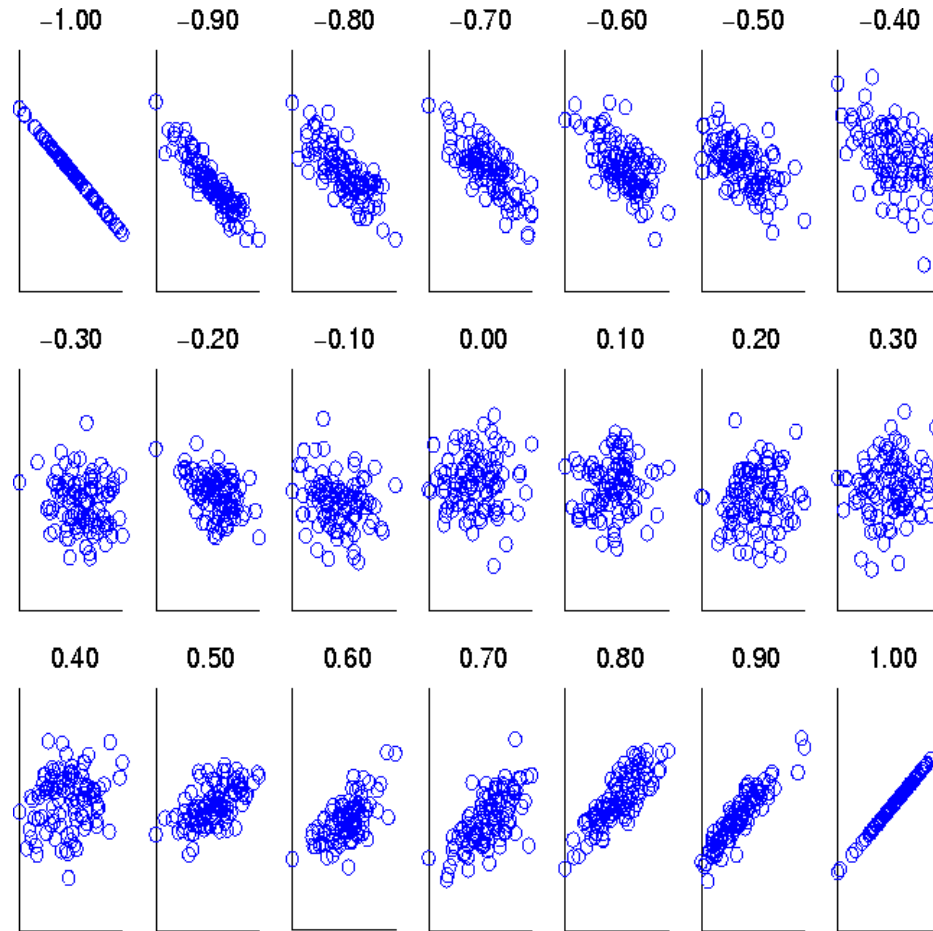
- Correlation measure the linear relationship between objects.
- To compute correlation, we standardize data objects, p and q , and then take the dot product.

$$p'_k = (p_k - \text{mean}(p)) / \text{std}(p)$$

$$q'_k = (q_k - \text{mean}(q)) / \text{std}(q)$$

$$\text{correlation}(p, q) = p' \bullet q'$$

Visually Evaluating Correlation



Scatter plots showing the similarity from -1 to 1

A General Approach for Combining Similarities

- Sometimes attributes are of many different types, but an overall similarity is needed.
 1. For the k^{th} attribute, compute a similarity, s_k , in the range $[0, 1]$.
 2. Define an indicator variable, δ_k , for the k_{th} attribute as follows:
$$\delta_k = \begin{cases} 0 & \text{if the } k^{th} \text{ attribute is a binary asymmetric attribute and both objects have} \\ & \text{a value of 0, or if one of the objects has a missing values for the } k^{th} \text{ attribute} \\ 1 & \text{otherwise} \end{cases}$$
 3. Compute the overall similarity between the two objects using the following formula:

$$similarity(p, q) = \frac{\sum_{k=1}^n \delta_k s_k}{\sum_{k=1}^n \delta_k}$$

Using Weights to Combine Similarities

- May not want to treat all attributes the same.
 - Use weights w_k which are between 0 and 1 and sum to 1.

$$\text{similarity}(p, q) = \frac{\sum_{k=1}^n w_k \delta_k s_k}{\sum_{k=1}^n \delta_k}$$

$$\text{distance}(p, q) = \left(\sum_{k=1}^n w_k |p_k - q_k|^r \right)^{1/r}$$

Problems with Selecting Initial Points

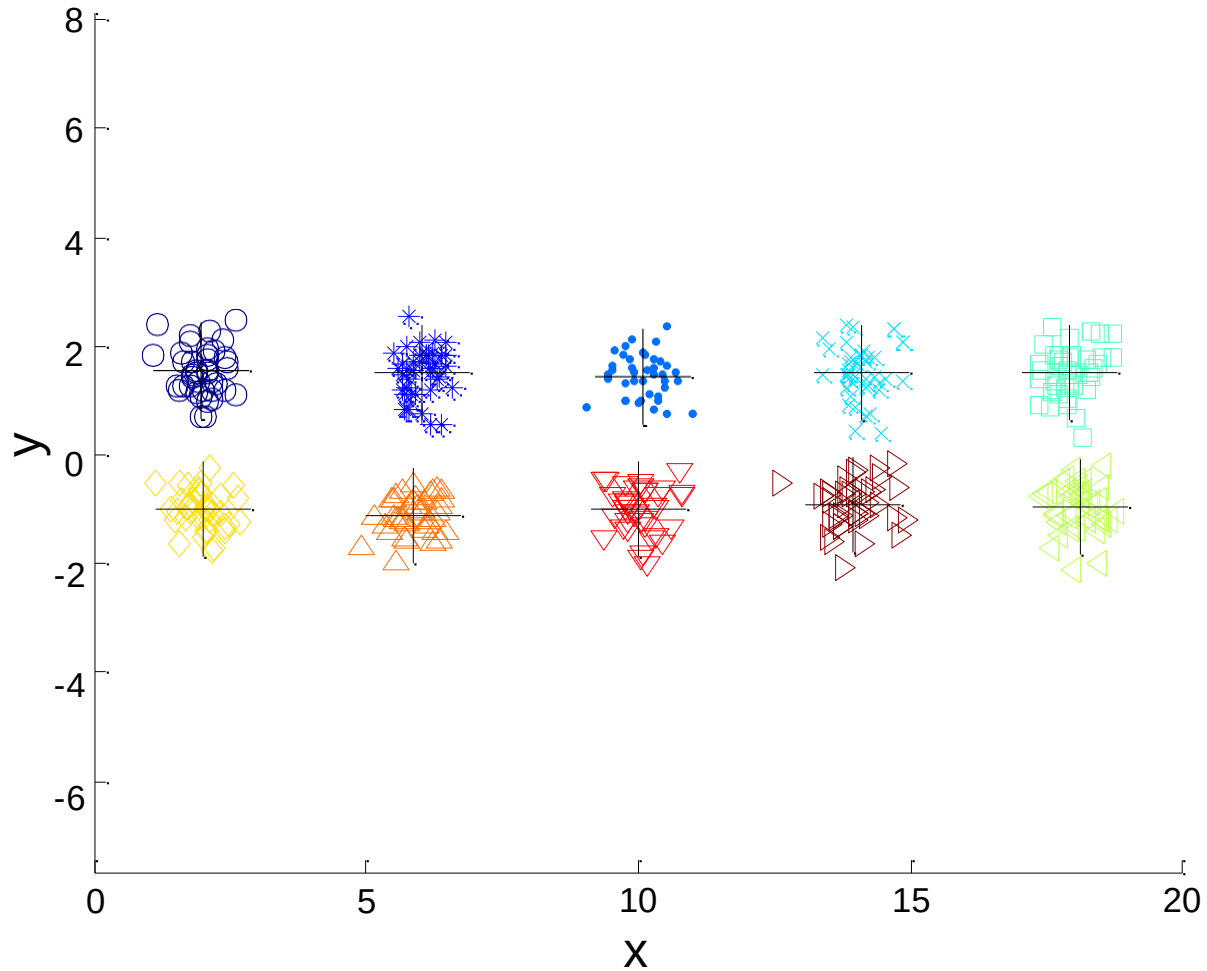
- If there are K 'real' clusters then the chance of selecting one centroid from each cluster is small.
 - Chance is relatively small when K is large
 - If clusters are the same size, n , then

$$P = \frac{\text{number of ways to select one centroid from each cluster}}{\text{number of ways to select } K \text{ centroids}} = \frac{K!n^K}{(Kn)^K} = \frac{K!}{K^K}$$

- For example, if $K = 10$, then probability = $10!/10^{10} = 0.00036$
- Sometimes the initial centroids will readjust themselves in 'right' way, and sometimes they don't
- Consider an example of five pairs of clusters

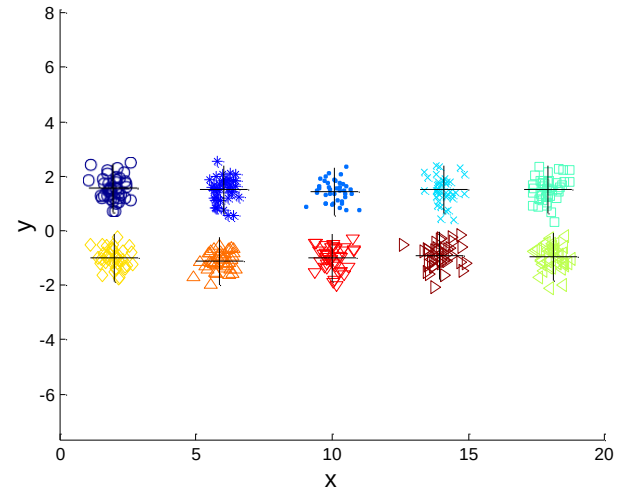
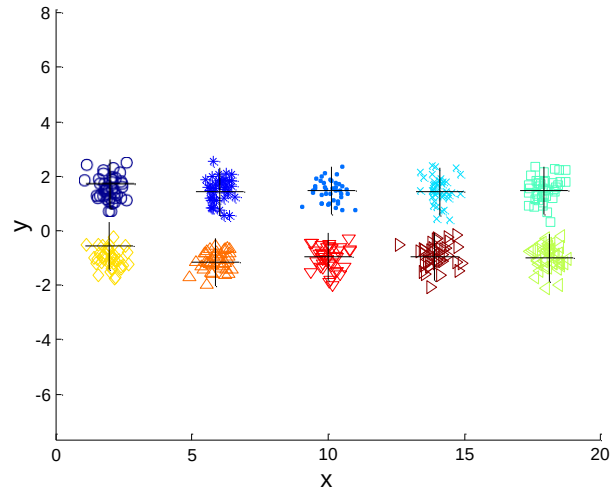
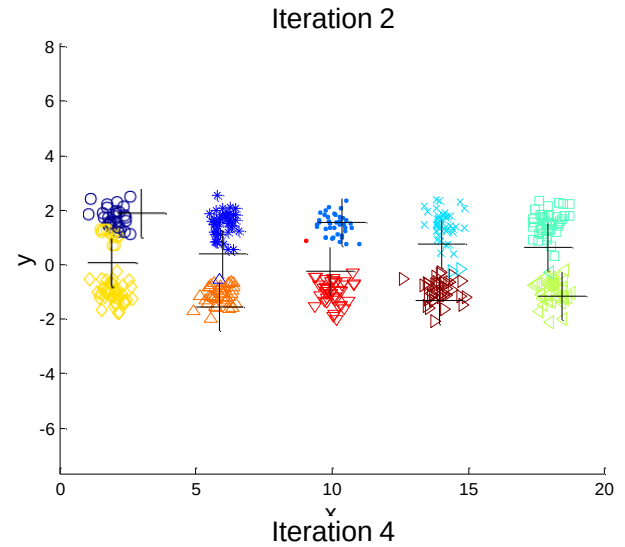
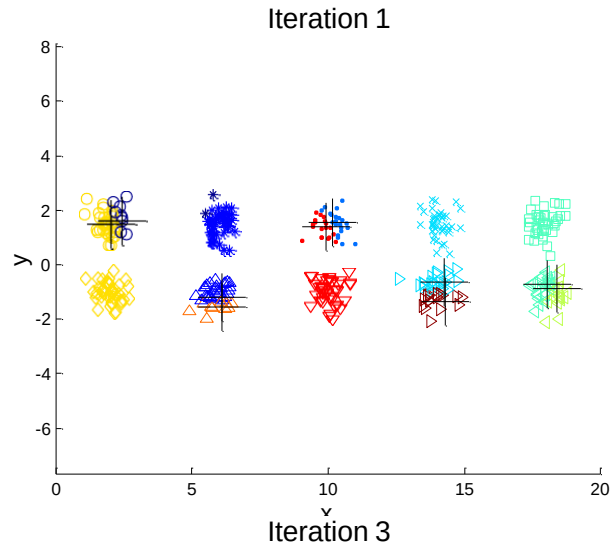
10 Clusters Example

Iteration 4



Starting with two initial centroids in one cluster of each pair of clusters

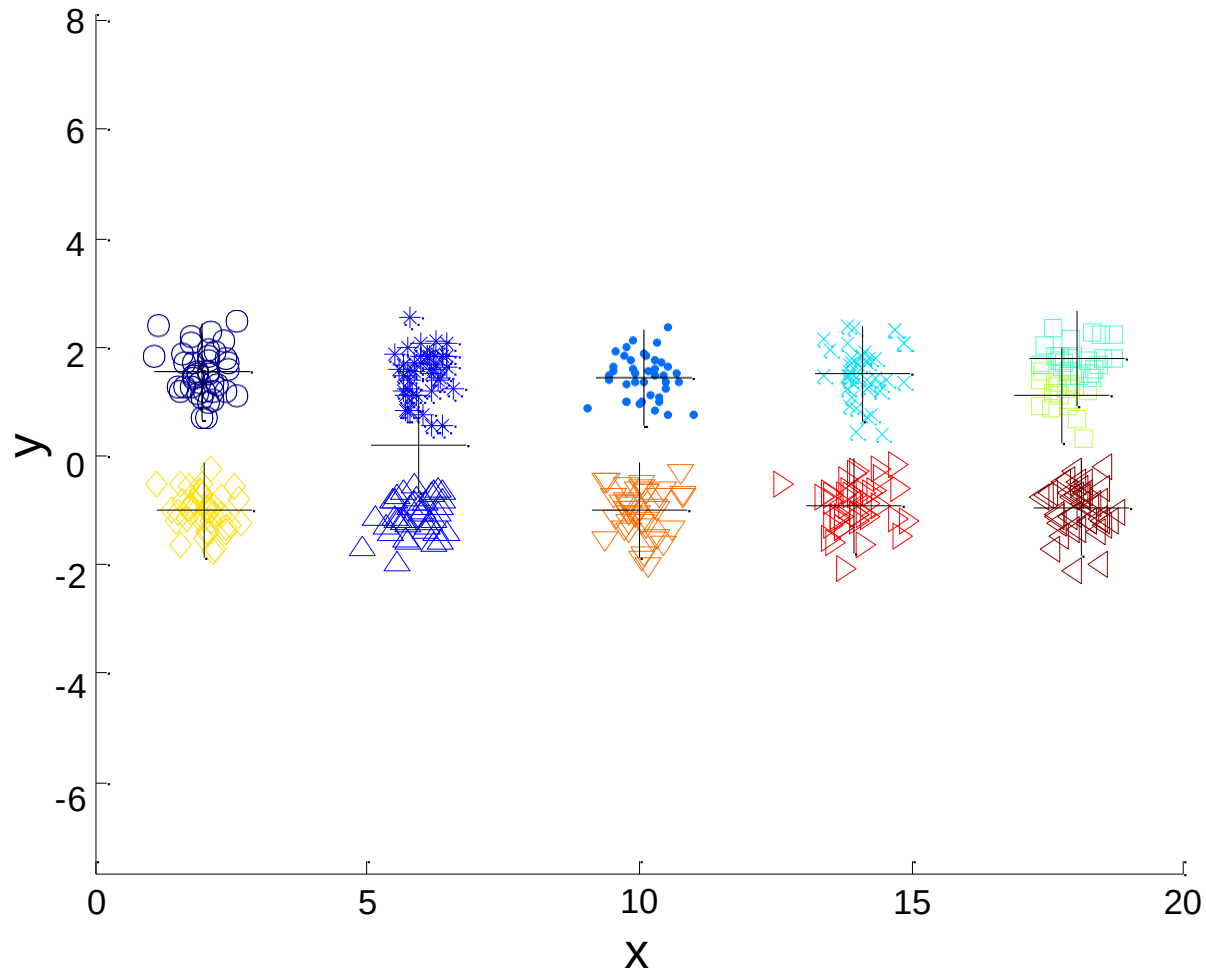
10 Clusters Example



Starting with two initial centroids in one cluster of each pair of clusters

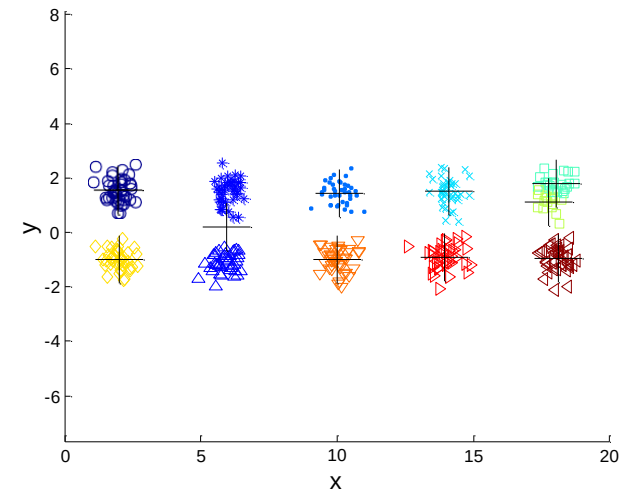
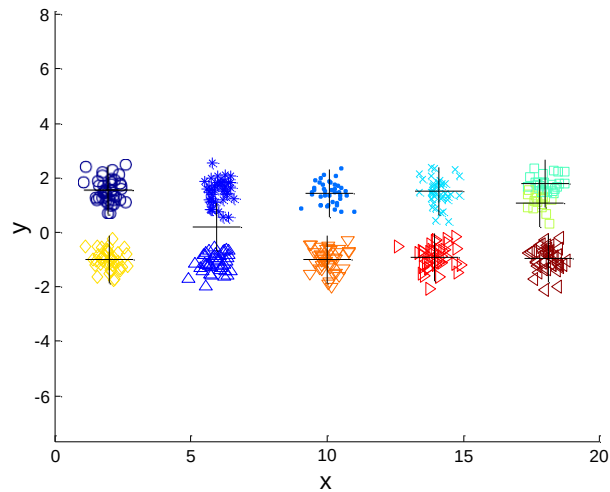
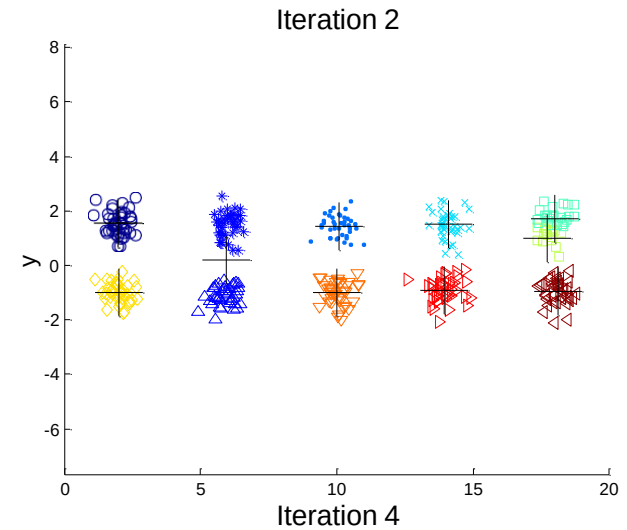
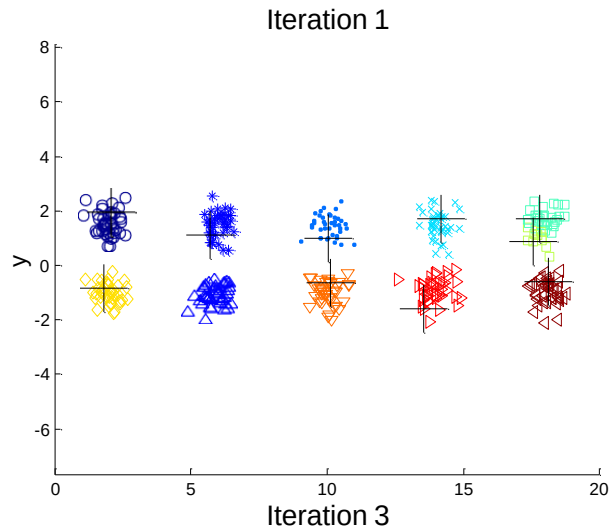
10 Clusters Example

Iteration 4



Starting with some pairs of clusters having three initial centroids, while other have only one.

10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Bisecting K-means
 - Not as susceptible to initialization issues
- Sample and use hierarchical clustering to determine initial Centroids
- Select more than K initial centroids and then select among these initial centroids
 - Select most widely separated
- Post-processing

Pre-processing and Post-processing

- Pre-processing
 - Normalize data so distance computations are fast.
 - Eliminate outliers
- Post-processing
 - Eliminate small clusters that may represent outliers
 - Split 'loose' clusters, i.e., clusters with relatively high SSE
 - Merge clusters that are 'close' and that have relatively low SSE

Bisecting K-means

- Bisecting K-means algorithm
 - Variant of K-means that can produce a partitional or a hierarchical clustering

```
1: Initialize the list of clusters to contain the cluster containing all points.
2: repeat
3:   Select a cluster from the list of clusters
4:   for  $i = 1$  to number_of_iterations do
5:     Bisect the selected cluster using basic K-means
6:   end for
7:   Add the two clusters from the bisection with the lowest SSE to the list of clusters.
8: until Until the list of clusters contains  $K$  clusters
```

Bisecting K-means

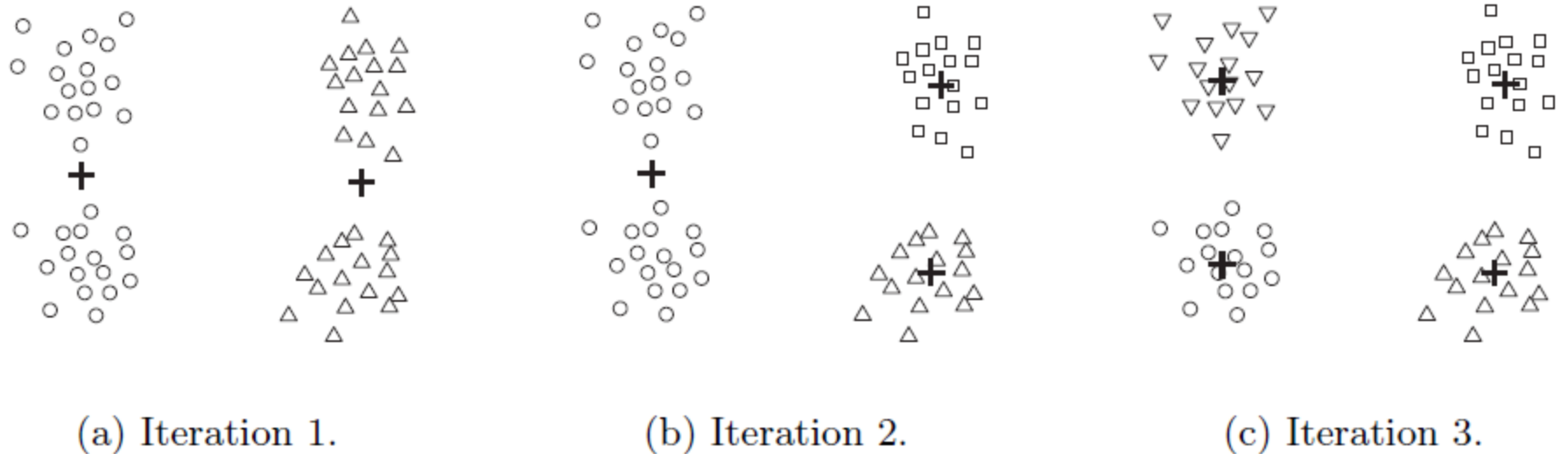
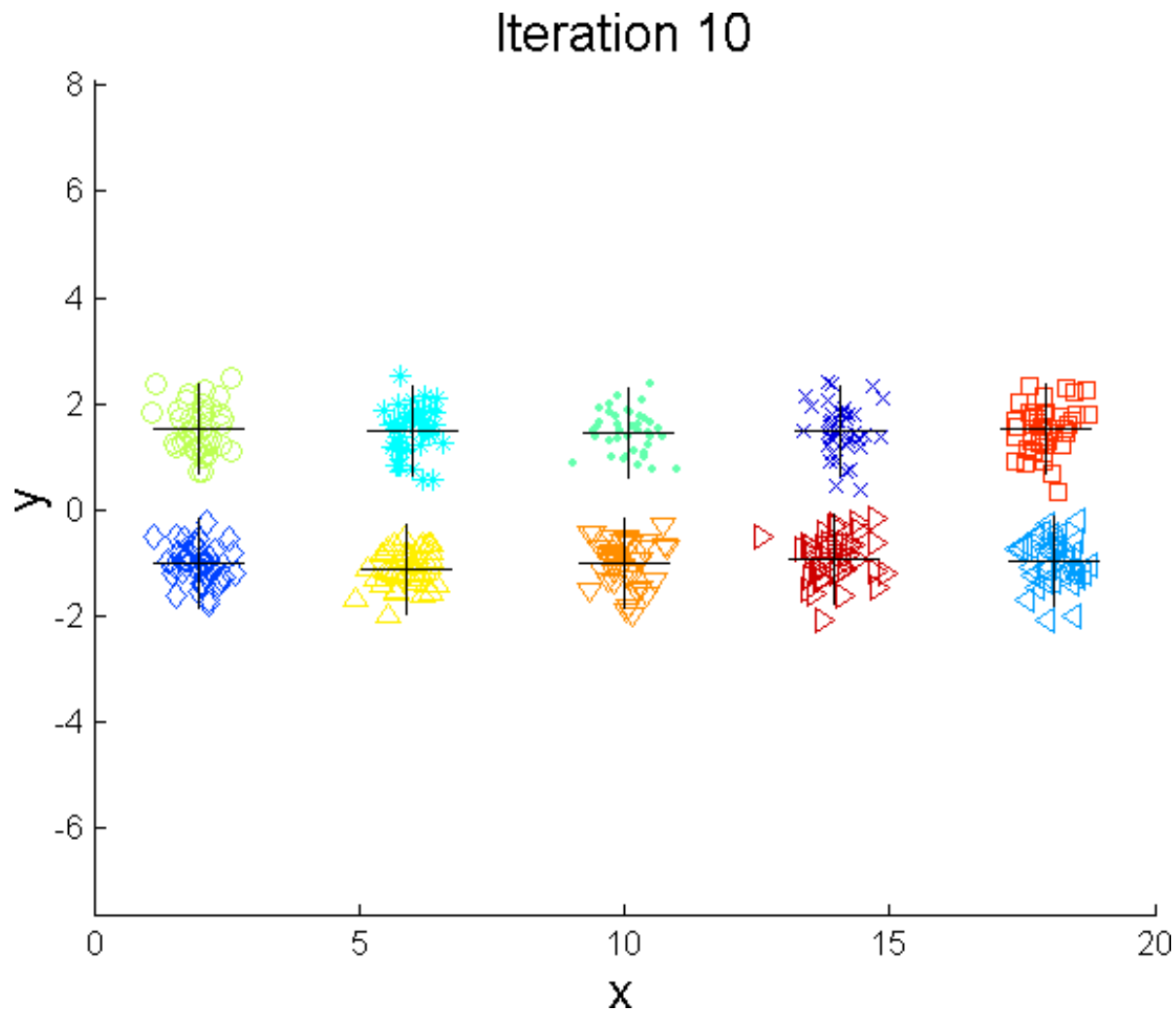


Figure 8.8. Bisecting K-means on the four clusters example.

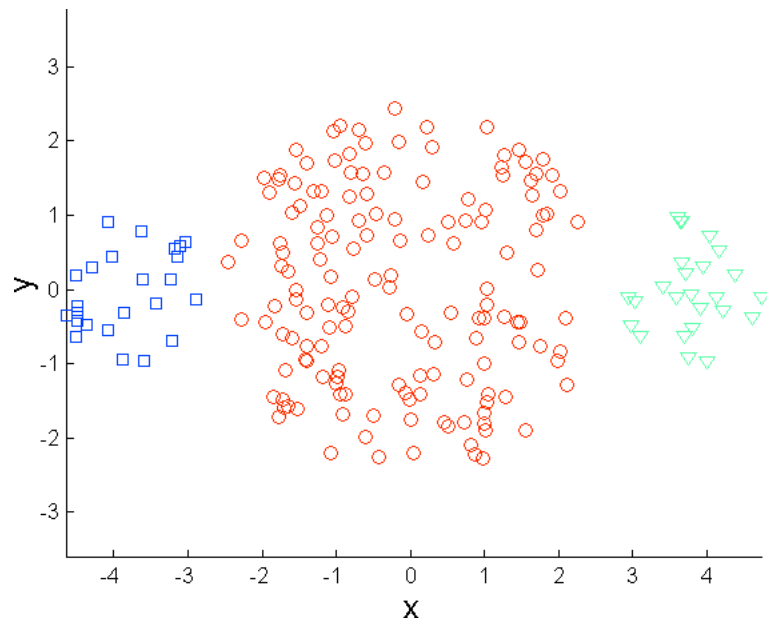
Bisecting K-means Example



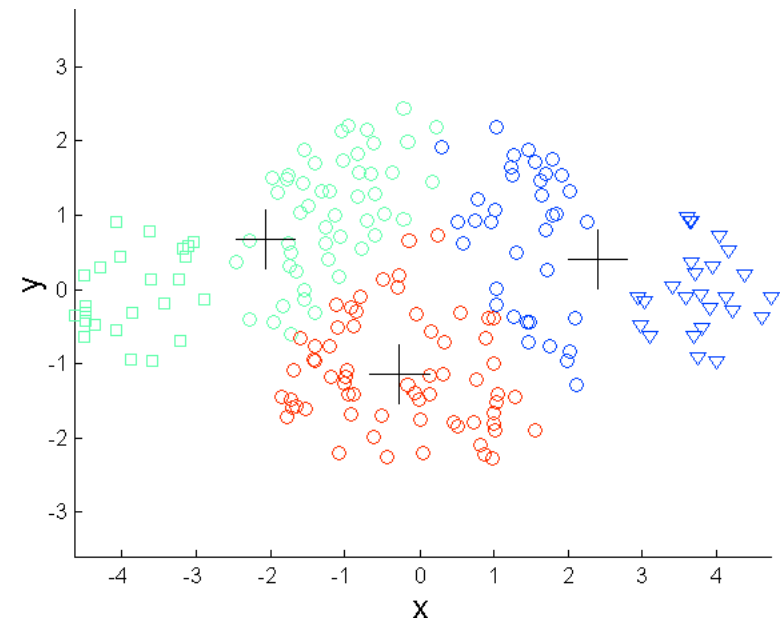
Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.
- One solution is to use many clusters.
 - Find parts of clusters, but need to put together.

Limitations of K-means: Differing Sizes

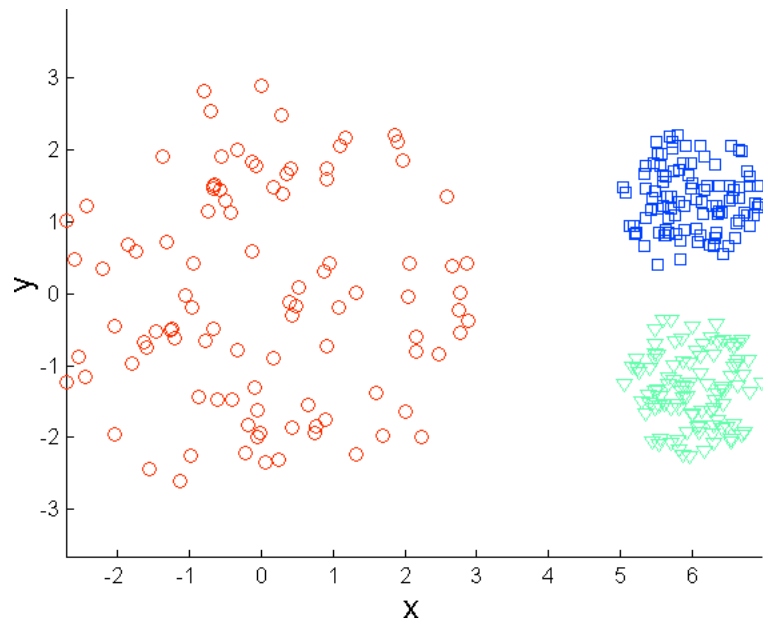


Original Points

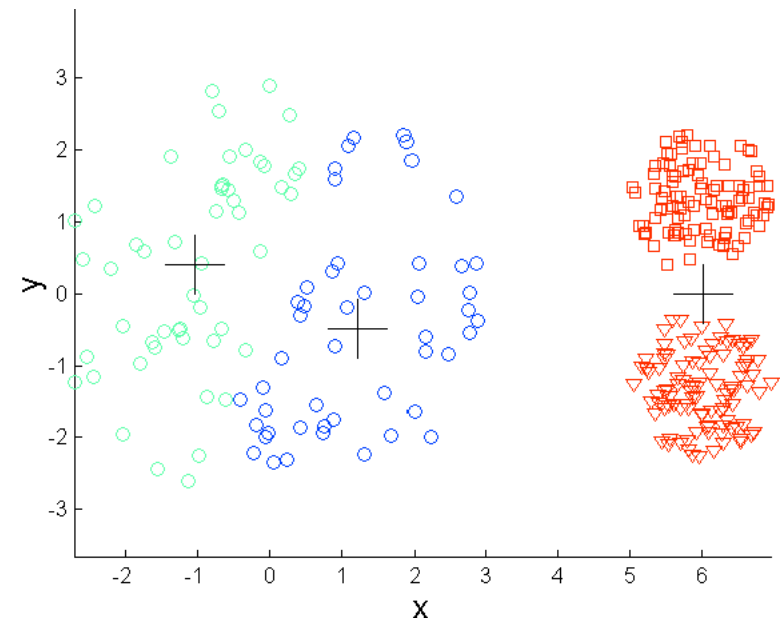


K-means Clusters

Limitations of K-means: Differing Density

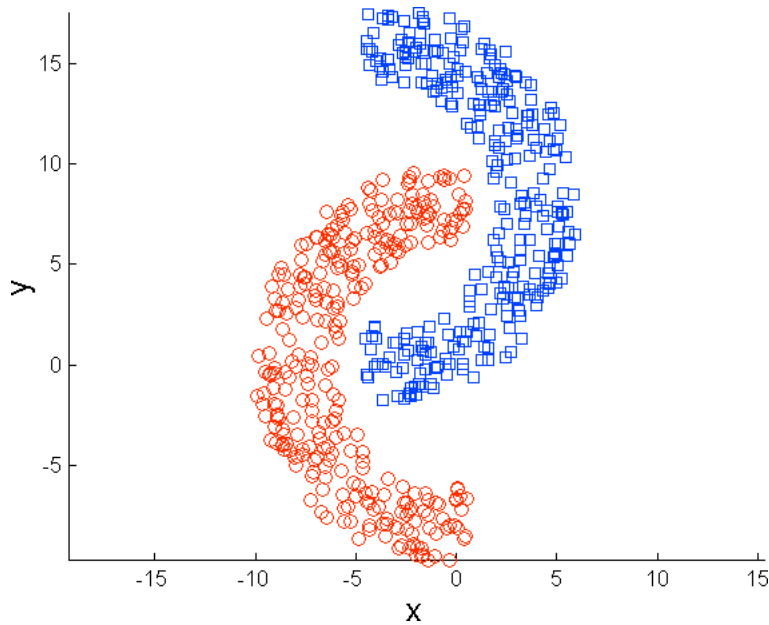


Original Points

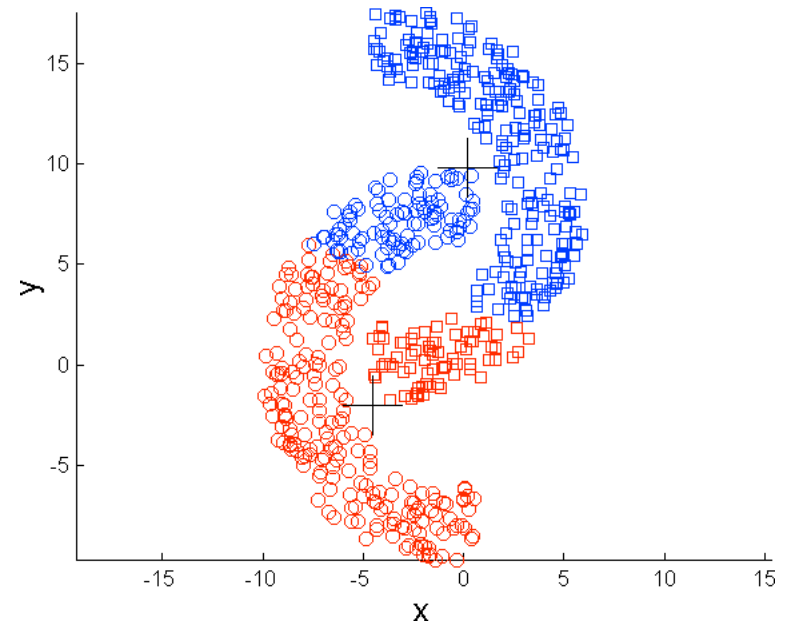


K-means Clusters

Limitations of K-means: Non-globular Shapes

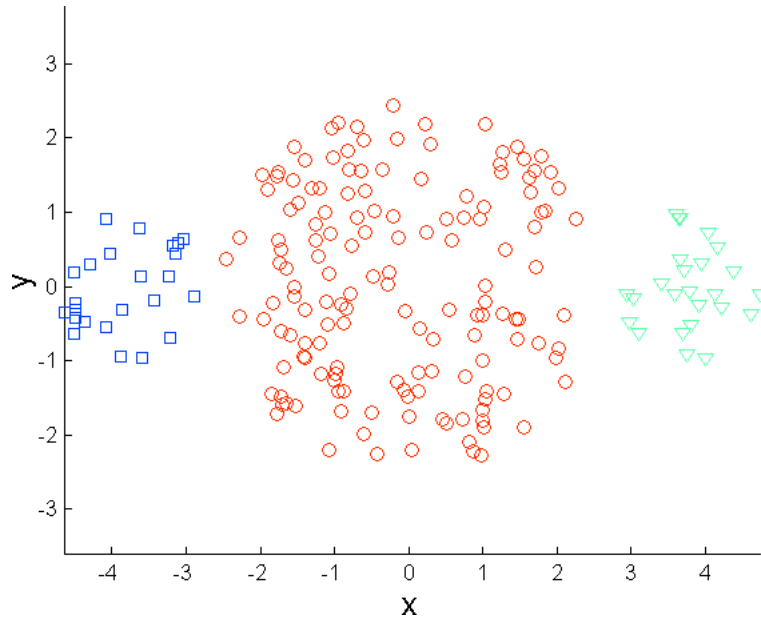


Original Points

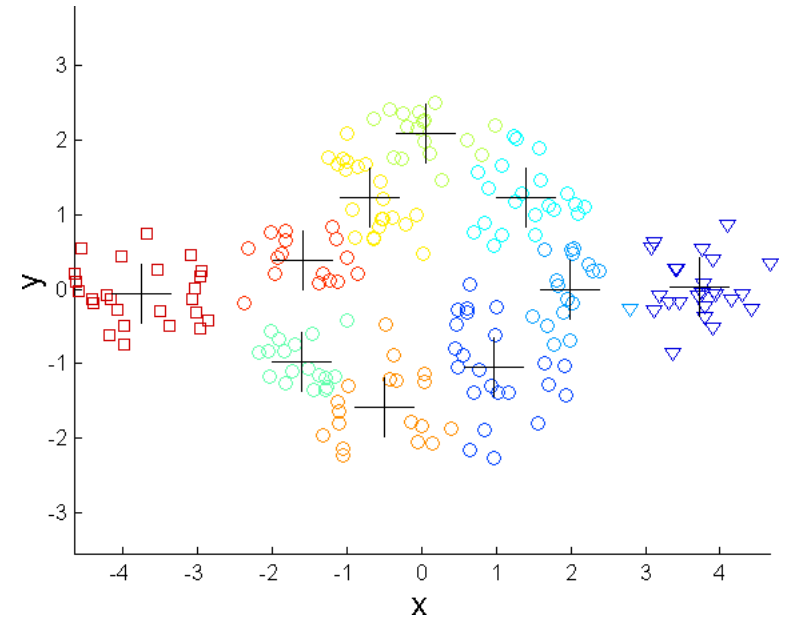


K-means Clusters

Overcoming K-means Limitations

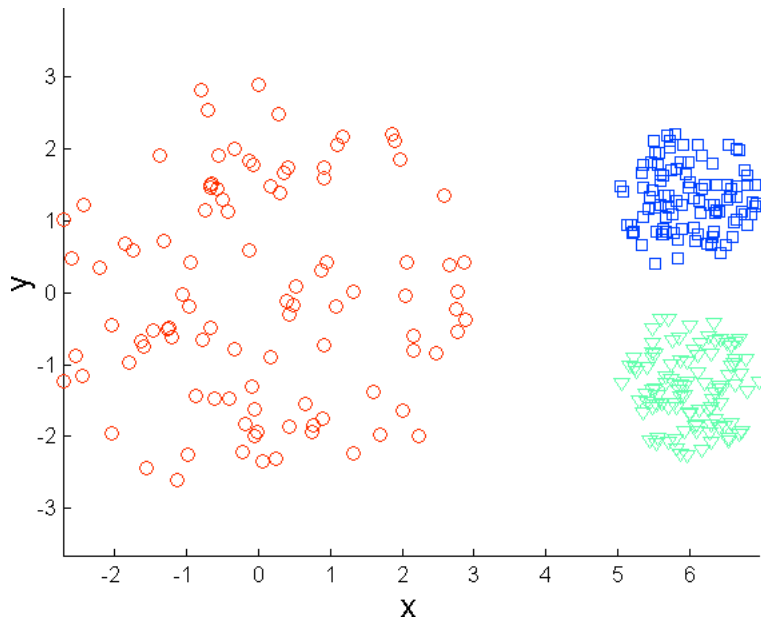


Original Points

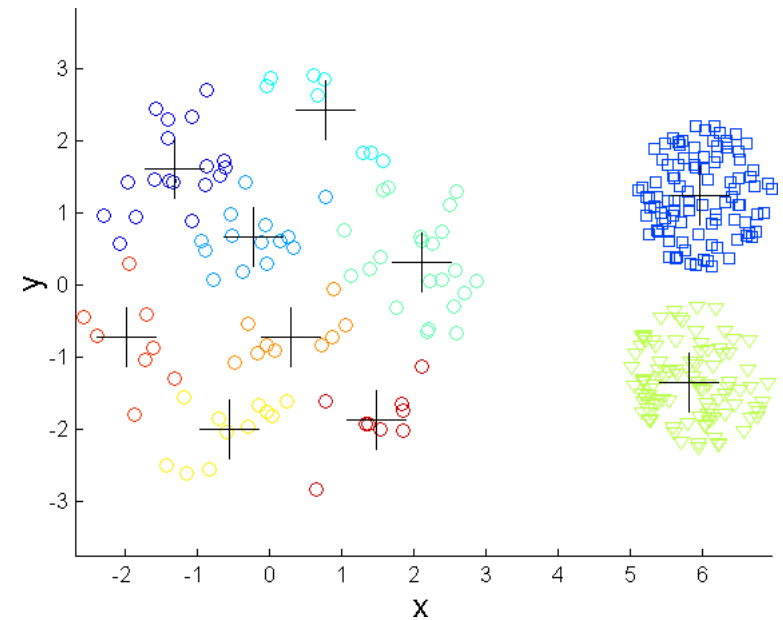


K-means Clusters

Overcoming K-means Limitations

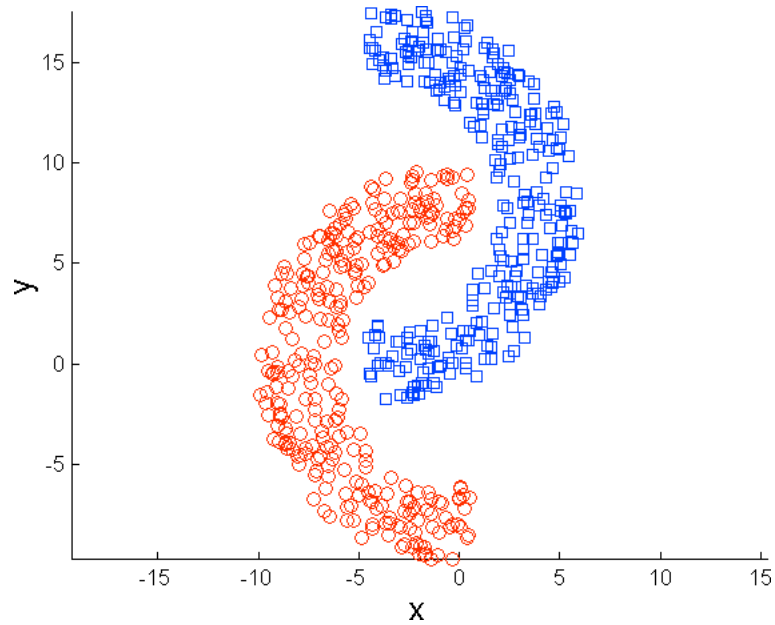


Original Points

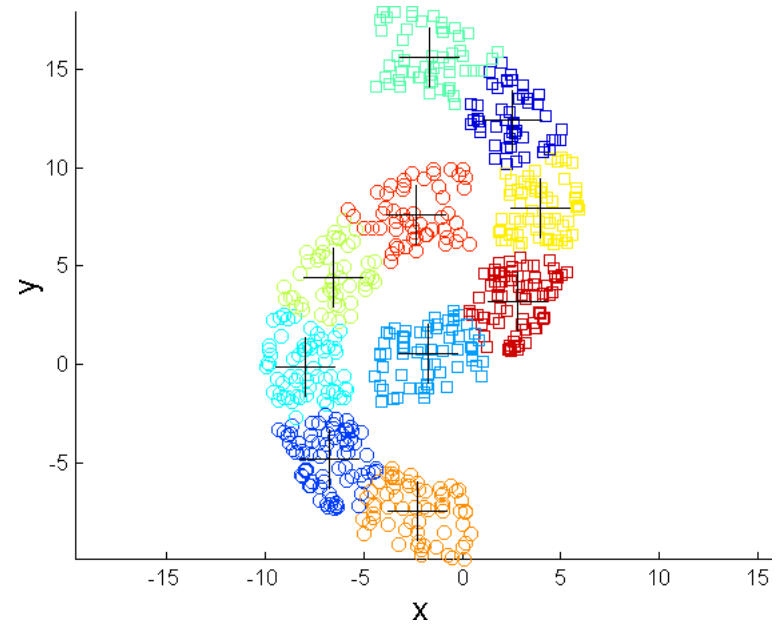


K-means Clusters

Overcoming K-means Limitations



Original Points



K-means Clusters

Thanks! ... *Questions?*